# Math Kangaroo Practice - Answer Key

For practice_set_2.pdf (Practice set for 2007 #14)

## Answer key & explanations

**2007 #14: C** The aquarium holds 3 × 3 × 3 = 27 small cubes in total. Counting the cubes already inside: the front layer has 1 cube, the middle layer has 2 + 1 = 3 cubes, and the back layer has 3 + 2 + 1 = 6 cubes, for a total of 1 + 3 + 6 = 10 cubes. Therefore, Daniella can put at most 27 - 10 = 17 more cubes into the aquarium.

**1.1: A** The aquarium holds 3 × 3 × 3 = 27 small cubes. Counting the cubes already inside: the front layer has 1 cube, the middle layer has 2 + 1 = 3 cubes, and the back layer has 3 + 2 + 2 = 7 cubes, for a total of 1 + 3 + 7 = 11 cubes. Therefore, Leo can put at most 27 - 11 = 16 more cubes into the aquarium.

**1.2: E** The aquarium holds 3 × 3 × 3 = 27 small cubes. Counting the cubes already inside: the front layer has 2 + 1 = 3 cubes, the middle layer has 2 + 2 = 4 cubes, and the back layer has 3 + 1 + 3 = 7 cubes, for a total of 3 + 4 + 7 = 14 cubes. Therefore, Maya can put at most 27 - 14 = 13 more cubes into the aquarium.

**1.3: B** The aquarium holds 3 × 3 × 3 = 27 small cubes. Counting the cubes already inside: the front layer has 1 + 1 = 2 cubes, the middle layer has 2 + 1 = 3 cubes, and the back layer has 3 + 2 + 1 = 6 cubes, for a total of 2 + 3 + 6 = 11 cubes. Lucas needs 27 - 11 = 16 more cubes. Since each box contains 3 cubes, 5 boxes provide 15 cubes (which is not enough), so Lucas needs at least 6 boxes (providing 18 cubes).